

Public Economics (ECON 131)

Section #2: Theoretical Tools of Public Finance

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1 Concept Review for Optimization

1.1 Utility Functions

- A **utility function** is a mathematical formulation of an individual's preferences. It is a function that translates consumption of goods into utility (you can think of this as happiness)
- Mathematically for two goods:

$$U = U(x_1, x_2)$$

- A common form of utility you will encounter is: **Cobb Douglas**¹

$$U(x_1, x_2) = x_1^\alpha x_2^\beta \text{ with } \alpha + \beta = 1$$

- Utility usually increases with level of consumption of a good²

1.2 Indifference Curves

- An **indifference curve** represents bundles of goods that make an individual equally well off, i.e. bundles for which utility is *constant*

- Mathematically:

$$\text{all } (x_1, x_2) \text{ where } U(x_1, x_2) = \bar{U}$$

↪ all bundles of (x_1, x_2) s.t. utility is equal $\bar{U}$

- Two main properties:

1. Consumers prefer higher indifference curves (more is better).
2. Indifference curves are usually downward sloping (if you take one good away from an individual, you have to give them some of the other good to keep them equally happy).

1.3 Marginal Utility

- Marginal utility is the additional utility obtained by consuming a little bit more of a good
- Mathematically, the marginal utility of x_1 is the derivative of utility with respect to x_1 , holding x_2 constant (i.e. the partial derivative)

$$MU_1 = \frac{\partial U}{\partial x_1}$$

- Graphically, the marginal utility of x_1 is the slope of the graph of U as a function of x_1 , at a given x_2

¹See your Section # 1 Notes if you need additional review of the useful properties of Cobb Douglas preferences.

²Can you think of a popular utility function where this does not hold?

1.4 Marginal Rate of Substitution (MRS)

- The MRS is the rate at which a consumer will trade x_2 for x_1
- Mathematically:

$$MRS_{1,2} = \frac{MU_1}{MU_2}$$

- Graphically: MRS is (minus) the slope of the indifference curve between x_1 and x_2

1.5 Budget Constraints

- The budget constraint is the mathematical representation of all the combinations of goods an individual can afford to buy if they spend their entire income:

$$p_1x_1 + p_2x_2 = y$$

if you only buy x_1 ,

if you only buy x_2

- Graphically, the line connects $\frac{y}{p_1}$ on the axis for good x_1 and $\frac{y}{p_2}$ on the axis for good x_2
- The slope of the budget constraint is $-\frac{p_1}{p_2}$

1.6 Example 1: Teaching Supplies

Sam is a kindergarten teacher in a cash-strapped public school. Each month, Sam has \$100 to buy new boxes of crayons (c) and games (g) for his students. Sam's utility is $U(c, g) = c^{1/2}g^{1/2}$. Suppose that crayons and games each cost \$2 at the store.

$$\hookrightarrow p_c = p_g = 2$$

$$\hookrightarrow \frac{MV_c}{MV_g} = \frac{p_c}{p_g}$$

- How many of each good does Sam buy? **Hint 1:** Try using the equimarginal principle to solve this problem. **Hint 2:** Transform the utility function using log (why can we do this?) $\rightarrow$ monotone

first, what's the B.C.? $100 = 2c + 2g$

log transformation $U(c, g) = \log(U(c, g))$
 $= \log(c^{1/2}g^{1/2})$
 $= \frac{1}{2}\log(c) + \frac{1}{2}\log(g)$

$$\frac{MV_c}{MV_g} = \frac{p_c}{p_g} \quad (\text{equimarginal principle})$$

$$MV_c = \frac{1}{2} \cdot \frac{1}{c} \quad \left[\text{if no log transformation } \frac{1}{2}c^{-1/2}g^{1/2} \dots \text{less clean} \right]$$

$$MV_g = \frac{1}{2} \cdot \frac{1}{g}$$

$$\text{then, } \frac{\frac{1}{2} \cdot \frac{1}{c}}{\frac{1}{2} \cdot \frac{1}{g}} = \frac{2}{2} = 1 \Rightarrow \frac{g}{c} = 1 \Rightarrow c^* = g^*$$

$$100 = 2c + 2c = 4c \Rightarrow c^* = 25 = g^*$$

- What is Sam's utility at (c^*, g^*) ?

plug these values into Sam's utility function $\Rightarrow U(25, 25) = (25)^{1/2}(25)^{1/2}$
 $= 25 \checkmark$
NOT the log transformation of it!

this is only to simplify the math!

2 The Effects of Price Changes $\rightarrow$ a net OR total effect = substitution + income

When the price of a good changes, this has two effects on an individual's consumption:

2.1 Substitution Effect

- The substitution effect is the change in consumption due to the change in relative prices between goods.
- This is a change along a given indifference curve

If the p_1 increases, what is the substitution effect on consumption of x_1 , if x_1 is a normal good?

if $p_1 \uparrow$, then $x_1 \downarrow$ via the substitution effect

$\rightarrow$ compare slopes

has to deal w/ opportunity cost
you will want to substitute to more x_2 and less x_1 (the more expensive good)

2.2 Income Effect

- The income effect is the change in consumption due to feeling poorer (if the price of the good increased) or richer (if the price decreased).
- This is a change from one indifference curve to another.

If the p_1 increases, what is the income effect on consumption of x_1 , if x_1 is a normal good?

if $p_1 \uparrow$, you feel poorer and if you feel poorer, consume less so $x_1 \downarrow$ through the income effect

Putting the together the two effects, what is the net effect on labor supply of increasing wage taxes?

$\uparrow$ wage tax $\downarrow$ wages

- you feel poorer, so via income effect you work more. labor supply $\uparrow$
- cheaper to consume leisure, so consume more of it. labor supply $\downarrow$

2.3 Teaching Supplies, continued

net effect is ambiguous
working is more expensive, consume less.

Let's revisit Sam's supply purchasing problem. Suppose there's a shortage of crayons and prices go up from \$2 per box to \$5. How does Sam reoptimize?

$$\frac{MV_c}{MV_g} = \frac{P_c}{P_g} \Rightarrow \frac{\frac{1}{2} \cdot \frac{1}{c}}{\frac{1}{2} \cdot \frac{1}{g}} = \frac{5}{2} \Rightarrow \frac{2}{c} = \frac{5}{g} \Rightarrow 2g = 5c$$

$$2g = 5(10)$$

$$g^* = 25$$

$$BC_{OLD}: 100 = 2g + 2c$$

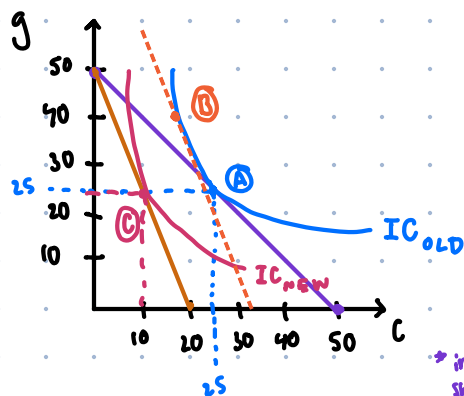
$$BC_{NEW}: 100 = 2g + 5c$$

$$\Rightarrow 100 = 5c + 5c = 10c$$

$$c^* = 10$$

Try drawing the old and new budget constraint and indifference curves to visually parse out the income effect and the substitution effect.

Try drawing the old and new budget constraint and indifference curves to visually parse out the income effect and the substitution effect.



→ indifference curves should be parallel to each other

$$BC_{OLD}: 100 = 2g + 2c$$

if $g=0$, then $100=2c$
 $c=50$

if $c=0$, then $100=2g$
 $g=50$

slope = -1

from the example, we know (25, 25) is the bundle that maximizes utility @ this B.C.

↳ that will serve as our point of tangency for indifference curve

$$BC_{NEW}: 100 = 2g + 5c$$

if $g=0$, then $100=5c$
 $c=20$

if $c=0$, then $100=2g$
 $g=50$

Slope = $-\frac{5}{2}$ steeper!

also from the example above, under this new B.C. the optimal bundle is (10, 25)

Ⓐ ⇒ Ⓑ is the substitution effect

Ⓑ ⇒ Ⓒ is the income effect

Ⓐ ⇒ Ⓒ is the total/net price effect

if $P_c \uparrow$

	c	g
substitution effect	↓	↑
income effect	↓	↓
total effect	↓	?

choose less of it!
→ which is costlier?
→ feel richer or poorer?

↳ ambiguous!

↳ however, after the math + figure, it cancels out w/ $g^*=25$

to acquire more of c, give up more of g than before

extra

some students asked how to mathematically find the tangency point ⑩
I do not foresee this being asked on a pretest/exam but have provided a solution below.

first: we know that point ⑩ and point ④ share the same utility of 25
[we derived this in example 1, part 2] b/c they are on the same indifference curve

therefore, we can conclude that $c^{1/2} g^{1/2} = 25$

second: the substitution effect forces us to make a choice based on the new prices. so, price of crayons is \$5 and price of games is \$2.
our income has changed, but it doesn't matter. assume income is y .

third: maximize the following

$$\max_{c, g} c^{1/2} g^{1/2} \quad \text{s.t.} \quad y = 5c + 2g \quad \text{and} \quad c^{1/2} g^{1/2} = 25$$

via equimarginal principle, we concluded above that

$$\frac{\frac{1}{2} \cdot \frac{1}{c}}{\frac{1}{2} \cdot \frac{1}{g}} = \frac{5}{2} \Rightarrow \frac{2}{c} = \frac{5}{g} \Rightarrow 2g = 5c$$
$$\Rightarrow g = \frac{5}{2}c$$

now,

$$c^{1/2} \cdot \left(\frac{5}{2}c\right)^{1/2} = 25$$

$$\Rightarrow c^{1/2} \cdot \left(\frac{5}{2}\right)^{1/2} \cdot c^{1/2} = 25$$

$$\Rightarrow 1.58c = 25$$

$$\Rightarrow c^* = 15.82 \quad \text{then} \quad g = \frac{5}{2}(15.82)$$

$$\Rightarrow g^* = 39.56$$

so, my graph is a little off but the intuition still holds!

3 Supply and Demand

3.1 Demand Curves

- The demand curve is the relationship between the price of the good and the quantity of the good
- The norm in economics is to plot quantity on the x-axis, and price on the y-axis.
- You can derive the demand curve from the utility maximization problem.
 - Solve the optimization problem *without* plugging numbers in for price. This will give you the quantity of each x as a function of p_1 and p_2
 - You can also do this graphically by solving the optimization problem for several specific values of price, and then plotting the relationship between quantity and price

Go back to your graph on the previous page, and use it to find two points on the demand curve for crayons.

3.2 Elasticity of Demand

- The elasticity of demand is the percent change in quantity demanded divided by the percent change in price.
- Mathematically, elasticity is given by:

$$\varepsilon_d = \frac{\partial q_d}{\partial p} \frac{p}{q_d} \approx \frac{\Delta q_d / q_d}{\Delta p / p}$$

$$\frac{\% \Delta \text{ in } Q_d}{\% \Delta \text{ in } p} = \frac{\frac{\Delta Q_d}{Q_D}}{\frac{\Delta P}{P}} \Rightarrow \frac{P}{Q_D} \cdot \frac{\partial Q_D}{\partial P}$$

- Conceptually useful because *unit free*
- Elasticity of demand is typically negative, because quantity demanded falls as price rises
- Demand is **perfectly inelastic** if the demand curve is vertical ($\varepsilon = 0$). The individual does not change the purchased quantity, no matter how much the price changes. ↳ not sensitive to Δ in price
- Demand is **perfectly elastic** if the demand curve is horizontal ($\varepsilon = -\infty$). Even the smallest price increase would cause the quantity demanded to fall to zero. ↳ sensitive to Δ in price

3.3 Supply Curves and Elasticity of Supply

- Each firm has a **Production Function**, which translates inputs into outputs.
- The **profits** of a firm are given by the difference between total revenue and total costs.
- Firms maximize profits. Mathematically, profit maximization is equivalent to setting marginal revenues equal to marginal costs. $MR = MC$ s.t. $\pi = MR - MC = 0$
- The supply curve defines the relationship between how much quantity of a good firms are willing to supply at each price.

- Solve the optimization problem *without* plugging numbers in for price. This will give you the quantity of each x as a function of p_1 and p_2
- You can also do this graphically by solving the optimization problem for several specific values of price, and then plotting the relationship between quantity and price

demand for crayon

$$y = p_c x_c + p_g x_g$$

$$\frac{MV_c}{MV_g} = \frac{p_c}{p_g}$$

$$\frac{1/x_c}{1/x_g} = \frac{p_c}{p_g} \Rightarrow$$

$$U(c, g) = c^{1/2} g^{1/2} + \frac{1}{2} \log(c) + \frac{1}{2} \log(g)$$

$$\frac{x_g}{x_c} = \frac{p_c}{p_g}$$

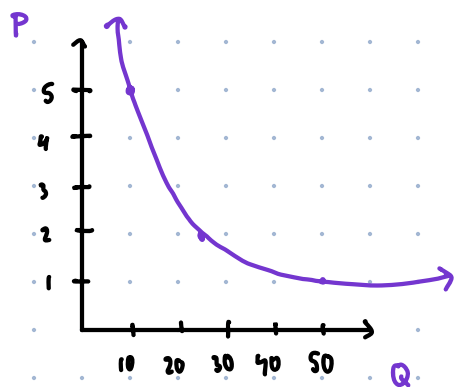
$$\Rightarrow x_g = x_c \cdot \frac{p_c}{p_g}$$

plug this into B.C.

$$y = p_c \cdot x_c + p_g \cdot \left[x_c \cdot \frac{p_c}{p_g} \right]$$

$$y = p_c \cdot x_c + x_c \cdot p_c = 2x_c p_c$$

$$\Rightarrow x_c = \frac{y}{2p_c}$$



old budget constraint

$$p_c = 2, \quad Q_c = 25$$

$$\frac{100}{2(2)} = 25$$

new budget constraint

$$p_c = 5, \quad Q_c = 10$$

$$\frac{100}{2(5)} = 10$$

$$\text{let } p_c = 1 \text{ \& } p_g = 2 \text{ \& } y = 100$$

$$x_c = \frac{100}{2(1)} = 50$$

- Supply curves are derived from profit maximization
- The elasticity of supply is the percent change in quantity supplied divided by the percent change in price.

$$\varepsilon_s = \frac{\partial q_s}{\partial p} \frac{p}{q_s} \approx \frac{\Delta q_s / q_s}{\Delta p / p}$$

4 Market Equilibrium

4.1 Definition

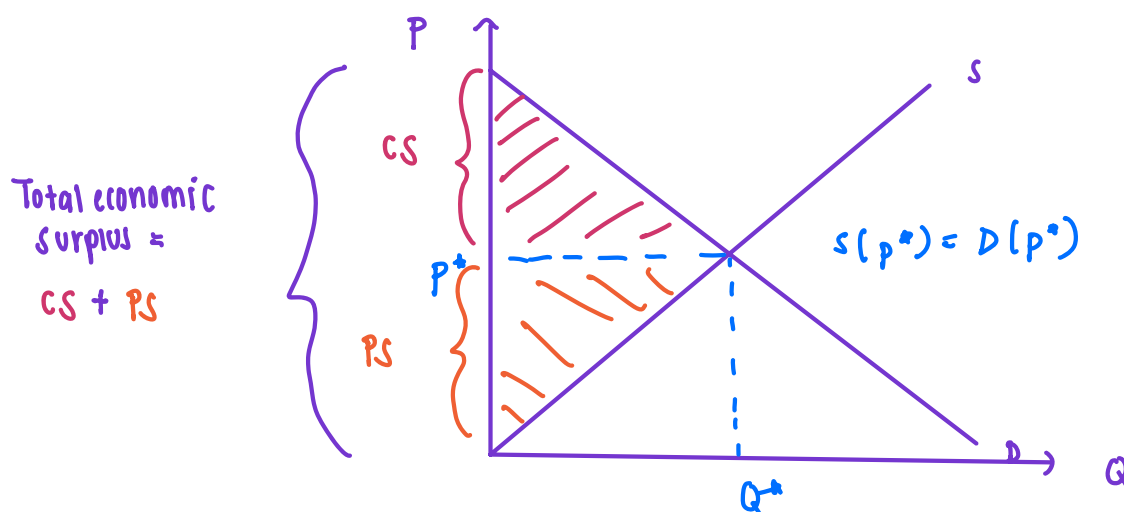
- Market equilibrium is the price p^* at which demand is equal to supply
- Mathematically, this is given by

$$D(p^*) = S(p^*)$$

- Graphically, this is the intersection of the demand curve and the supply curve.

4.2 Economic Surplus

- **Consumer surplus** is the benefit consumers derive from consuming the good when their valuation of the good is higher than the price they paid. It is the area below the demand curve and above the horizontal line denoting market price.
- **Producer surplus** is the benefit producers derive from selling a good when the market price is higher than what they would have been willing to sell it for. It is the area above the supply curve and below the horizontal line denoting market price.
- **Total economic surplus** is the sum of consumer surplus and producer surplus. It is the area above the supply curve and below the demand curve:



5 Social Welfare

5.1 The Fundamental Theorems of Welfare Economics

- The **First Fundamental Theorem of Welfare Economics** states under 1) no externalities; 2) perfect competition; and 3) perfect information: the competitive equilibrium (i.e. market; where supply meets demand) is **Pareto Efficient**:
 - **Pareto Efficiency**: changes to equilibrium makes someone worse-off (this is a *weak* notion of efficiency!)
 - Any price that deviates from the equilibrium generates **deadweight loss**, which is a reduction in total economic surplus.
 - These conditions are *stringent*, and mostly violated in practice: markets with externalities, imperfect competition, transaction costs, irrational agents or imperfect information may not yield efficient results.
 - This does not mean that markets cannot be a useful tool for allocating resources!
- The **Second Fundamental Theorem of Welfare Economics** states that any Pareto Efficient allocation can be achieved through a tax and transfer system where individuals do not respond to taxation.
 - In theory: lump-sum taxes and transfers based on individual ability have this feature
 - In practice, the government cannot perfectly measure ability, and other types of taxes change prices/individuals' choices (we saw this before with labor taxes). Redistribution will distort the market equilibrium away from the Pareto optimum – we rarely attain the "first-best" equilibrium.
 - This insight leads to the **equity-efficiency trade off**.

5.2 Social Welfare Functions

A social welfare function (SWF) combines the utility functions of all individuals into an aggregate utility function. The way that utilities are combined determines how equity is valued. We usually write this as: $W(u_1, u_2, \dots)$

There are many different types of SWFs, depending on what we value as a society. Two famous examples:

- **Utilitarian**: maximize the sum of individual utilities.
- **Rawlsian**: maximize the well-being of the worst-off member.